

CRM Database Design

Based on SRS v1.0

Legend: Column Header Primary Key Foreign Key

1. users

Description: Stores CRM system users who can be assigned Leads, Activities, Opportunities, Tickets, and other responsibilities.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
user_id	BIGINT	NO	AUTO_INCREMENT	Unique identifier for the user	PK	—
role_id	BIGINT	NO	—	Role assigned to the user	FK	roles.role_id
first_name	VARCHAR(100)	NO	—	User's first name	—	—
last_name	VARCHAR(100)	NO	—	User's last name	—	—
email	VARCHAR(255)	NO	—	User login email address	UNIQUE	—
password_hash	VARCHAR(255)	NO	—	Hashed and salted password	—	—
phone	VARCHAR(50)	YES	NULL	User phone number	—	—
status	VARCHAR(20)	NO	active	User account status	active/deactivated	—
last_login_at	DATETIME	YES	NULL	Date and time of last successful login	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	User creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—
deactivated_at	DATETIME	YES	NULL	Date/time the user was deactivated	—	—

The SRS requires administrators to create/deactivate/manage users and specifies soft deactivation rather than hard deletion.

2. roles

Description: Stores the four CRM roles defined by SRS v1.0.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
role_id	BIGINT	NO	AUTO_INCREMENT	Unique role identifier	PK	—
role_name	VARCHAR(50)	NO	—	Role name	UNIQUE	—
description	VARCHAR(500)	YES	NULL	Description of role responsibilities	—	—
role_tier	INT	YES	NULL	Relative role seniority	—	—

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
is_active	BOOLEAN	NO	TRUE	Indicates whether role is active	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

Initial roles: CRM Manager, Business Development Officer, Support Agent, System Administrator.

3. leads

Description: Stores unqualified prospective relationships captured from different sources.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
lead_id	BIGINT	NO	AUTO_INCREMENT	Unique Lead identifier	PK	—
owner_user_id	BIGINT	YES	NULL	Current user responsible for the Lead	FK	users.user_id
name	VARCHAR(255)	NO	—	Lead name	—	—
organization	VARCHAR(255)	YES	NULL	Lead organization name	—	—
email	VARCHAR(255)	YES	NULL	Lead email address	—	—
phone	VARCHAR(50)	YES	NULL	Lead phone number	—	—
source_id	BIGINT	YES	NULL	Source from which Lead originated	FK	lead_sources.source_id
category_id	BIGINT	YES	NULL	Lead category	FK	reference_categories.category_id
status	VARCHAR(30)	NO	new	Current Lead status	new/contacted/qualified/rejected	—
notes	TEXT	YES	NULL	Additional Lead notes	—	—
converted_at	DATETIME	YES	NULL	Date/time Lead was converted	—	—
deleted_at	DATETIME	YES	NULL	Soft deletion timestamp	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

The Lead requirements include creation, viewing, updating, deletion, assignment, status tracking, and conversion.

4. lead_sources

Description: Stores configurable Lead source values.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
source_id	BIGINT	NO	AUTO_INCREMENT	Unique source identifier	PK	—
source_name	VARCHAR(100)	NO	—	Lead source name	UNIQUE	—
description	VARCHAR(255)	YES	NULL	Description of source	—	—

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
is_active	BOOLEAN	NO	TRUE	Whether source is available	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

The SRS lists Lead sources as reference/master data owned by the System Administrator.

5. accounts

Description: Stores organizations such as startups, investors, companies, government organizations, and partners.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
account_id	BIGINT	NO	AUTO_INCREMENT	Unique Account identifier	PK	—
owner_user_id	BIGINT	YES	NULL	Current Account owner	FK	users.user_id
organization_name	VARCHAR(255)	NO	—	Organization name	—	—
account_type_id	BIGINT	NO	—	Organization category	FK	reference_categories.category_id
industry	VARCHAR(150)	YES	NULL	Industry of organization	—	—
location	VARCHAR(255)	YES	NULL	Organization location	—	—
contact_information	TEXT	YES	NULL	General account contact information	—	—
engagement_history	TEXT	YES	NULL	Summary of engagement history	—	—
deleted_at	DATETIME	YES	NULL	Soft deletion timestamp	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

6. contacts

Description: Stores individual people associated with Accounts.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
contact_id	BIGINT	NO	AUTO_INCREMENT	Unique Contact identifier	PK	—
account_id	BIGINT	NO	—	Account to which Contact belongs	FK	accounts.account_id
owner_user_id	BIGINT	YES	NULL	Current Contact owner	FK	users.user_id
first_name	VARCHAR(100)	NO	—	Contact first name	—	—
last_name	VARCHAR(100)	NO	—	Contact last name	—	—
role_title	VARCHAR(150)	YES	NULL	Contact's role/title	—	—
phone	VARCHAR(50)	YES	NULL	Contact phone number	—	—

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
email	VARCHAR(255)	YES	NULL	Contact email address	—	—
is_primary	BOOLEAN	NO	FALSE	Indicates Primary Contact for Account	—	—
deleted_at	DATETIME	YES	NULL	Soft deletion timestamp	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

SRS v1.0 defines one Account per Contact in the baseline and introduces a Primary Contact rule.

7. opportunities

Description: Stores qualified commercial/business engagements derived from Leads.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
opportunity_id	BIGINT	NO	AUTO_INCREMENT	Unique Opportunity identifier	PK	—
lead_id	BIGINT	YES	NULL	Lead from which Opportunity originated	FK	leads.lead_id
account_id	BIGINT	NO	—	Account associated with Opportunity	FK	accounts.account_id
owner_user_id	BIGINT	YES	NULL	Current Opportunity owner	FK	users.user_id
opportunity_type_id	BIGINT	NO	—	Opportunity type	FK	opportunity_types.opportunity_type_id
stage_id	BIGINT	NO	—	Current pipeline stage	FK	pipeline_stages.stage_id
probability	DECIMAL(5,2)	YES	NULL	Probability of closing	—	—
value	DECIMAL(18,2)	YES	NULL	Opportunity value	—	—
expected_close_date	DATE	YES	NULL	Expected closing date	—	—
notes	TEXT	YES	NULL	Opportunity notes	—	—
status	VARCHAR(30)	NO	open	Opportunity lifecycle status	open/closed_won/closed_lost	—
closed_at	DATETIME	YES	NULL	Date/time Opportunity was closed	—	—
deleted_at	DATETIME	YES	NULL	Soft deletion timestamp	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

The SRS requires investment, partnership, service, and program enrollment Opportunity types, configurable stages, probability, value, expected closing date, stage history, and Closed Won/Lost.

8. opportunity_types

Description: Stores configurable Opportunity types.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
opportunity_type_id	BIGINT	NO	AUTO_INCREMENT	Unique Opportunity type identifier	PK	—
type_name	VARCHAR(100)	NO	—	Opportunity type name	UNIQUE	—
description	VARCHAR(255)	YES	NULL	Type description	—	—
is_active	BOOLEAN	NO	TRUE	Whether type is active	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

9. pipeline_stages

Description: Stores configurable Opportunity pipeline stages.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
stage_id	BIGINT	NO	AUTO_INCREMENT	Unique stage identifier	PK	—
stage_name	VARCHAR(100)	NO	—	Pipeline stage name	UNIQUE	—
display_order	INT	NO	—	Stage order in pipeline	—	—
is_closed_won	BOOLEAN	NO	FALSE	Indicates Closed Won stage	—	—
is_closed_lost	BOOLEAN	NO	FALSE	Indicates Closed Lost stage	—	—
is_active	BOOLEAN	NO	TRUE	Whether stage is active	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

Default stages are defined as New → Contacted → Qualified → Proposal/Negotiation → Closed Won/Lost.

10. opportunity_stage_history

Description: Records historical changes to Opportunity pipeline stages.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
history_id	BIGINT	NO	AUTO_INCREMENT	Unique history record	PK	—
opportunity_id	BIGINT	NO	—	Opportunity whose stage changed	FK	opportunities.opportunity_id
from_stage_id	BIGINT	YES	NULL	Previous stage	FK	pipeline_stages.stage_id
to_stage_id	BIGINT	NO	—	New stage	FK	pipeline_stages.stage_id
changed_by_user_id	BIGINT	NO	—	User who changed stage	FK	users.user_id
changed_at	DATETIME	NO	CURRENT_TIMESTAMP	Date/time of change	—	—
notes	TEXT	YES	NULL	Optional stage-change notes	—	—

11. activities

Description: Stores calls, meetings, emails, and tasks associated with CRM records.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
activity_id	BIGINT	NO	AUTO_INCREMENT	Unique Activity identifier	PK	—
assigned_user_id	BIGINT	YES	NULL	User responsible for Activity	FK	users.user_id
lead_id	BIGINT	YES	NULL	Related Lead	FK	leads.lead_id
account_id	BIGINT	YES	NULL	Related Account	FK	accounts.account_id
contact_id	BIGINT	YES	NULL	Related Contact	FK	contacts.contact_id
opportunity_id	BIGINT	YES	NULL	Related Opportunity	FK	opportunities.opportunity_id
activity_type	VARCHAR(30)	NO	—	Call, meeting, email, or task	—	—
subject	VARCHAR(255)	NO	—	Activity subject	—	—
description	TEXT	YES	NULL	Activity details	—	—
scheduled_at	DATETIME	YES	NULL	Scheduled date/time	—	—
completed_at	DATETIME	YES	NULL	Completion date/time	—	—
reminder_at	DATETIME	YES	NULL	Reminder date/time	—	—
status	VARCHAR(30)	NO	planned	Activity status	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

12. communications

Description: Stores communication history including emails, messages, call logs, and manually entered communication notes.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
communication_id	BIGINT	NO	AUTO_INCREMENT	Unique communication identifier	PK	—
account_id	BIGINT	YES	NULL	Related Account	FK	accounts.account_id
contact_id	BIGINT	YES	NULL	Related Contact	FK	contacts.contact_id
lead_id	BIGINT	YES	NULL	Related Lead	FK	leads.lead_id
opportunity_id	BIGINT	YES	NULL	Related Opportunity	FK	opportunities.opportunity_id
activity_id	BIGINT	YES	NULL	Related Activity	FK	activities.activity_id
created_by_user_id	BIGINT	NO	—	User who recorded communication	FK	users.user_id
communication_type	VARCHAR(30)	NO	—	Email, message, call, or note	—	—
subject	VARCHAR(255)	YES	NULL	Communication subject	—	—

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
content	TEXT	YES	NULL	Communication content/note	—	—
communication_at	DATETIME	NO	CURRENT_TIMESTAMP	Communication date/time	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Record creation timestamp	—	—

External email integration is explicitly deferred in v1.0, so this table supports communication history/logging, not necessarily live email synchronization.

13. tickets

Description: Stores customer support/service requests.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
ticket_id	BIGINT	NO	AUTO_INCREMENT	Unique Ticket identifier	PK	—
account_id	BIGINT	YES	NULL	Account associated with Ticket	FK	accounts.account_id
contact_id	BIGINT	YES	NULL	Contact associated with Ticket	FK	contacts.contact_id
assignee_user_id	BIGINT	YES	NULL	Support Agent assigned to Ticket	FK	users.user_id
category_id	BIGINT	NO	—	Ticket category	FK	ticket_categories.category_id
priority	VARCHAR(20)	NO	Medium	Ticket priority	Urgent/High/Medium/Low	—
status	VARCHAR(30)	NO	open	Ticket status	open/in progress/resolved/closed	—
subject	VARCHAR(255)	NO	—	Ticket subject	—	—
description	TEXT	YES	NULL	Ticket description	—	—
last_activity_at	DATETIME	NO	CURRENT_TIMESTAMP	Timestamp of latest Ticket activity	—	—
inactivity_duration_seconds	BIGINT	YES	NULL	Calculated inactivity duration	—	—
escalated	BOOLEAN	NO	FALSE	Current escalation flag	—	—
escalated_at	DATETIME	YES	NULL	Time current escalation occurred	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—
closed_at	DATETIME	YES	NULL	Ticket closure timestamp	—	—
deleted_at	DATETIME	YES	NULL	Soft deletion timestamp	—	—

The SRS specifically defines escalation as a flag independent of Ticket status, rather than another status.

14. ticket_categories

Description: Stores configurable Ticket categories.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
category_id	BIGINT	NO	AUTO_INCREMENT	Unique category identifier	PK	—
category_name	VARCHAR(100)	NO	—	Ticket category name	UNIQUE	—
description	VARCHAR(255)	YES	NULL	Category description	—	—
is_active	BOOLEAN	NO	TRUE	Whether category is active	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

Default categories: Technical, Service Request, General Inquiry.

15. escalation_policies

Description: Stores configurable Ticket inactivity thresholds by Priority.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
escalation_policy_id	BIGINT	NO	AUTO_INCREMENT	Unique policy identifier	PK	—
priority	VARCHAR(20)	NO	—	Ticket priority to which policy applies	Urgent/High/Medium/Low	—
reminder_threshold_minutes	INT	NO	—	Inactivity duration before reminder	Must be less than escalation threshold	—
escalation_threshold_minutes	INT	NO	—	Inactivity duration before escalation	—	—
time_basis	VARCHAR(30)	NO	calendar	Time calculation basis	calendar/business_hours	—
is_active	BOOLEAN	NO	TRUE	Whether policy is active	—	—
created_by_user_id	BIGINT	NO	—	Administrator who created policy	FK	users.user_id
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

The SRS requires configurable reminder and escalation thresholds and requires the reminder threshold to be strictly less than the escalation threshold.

16. escalation_recipients

Description: Stores users/roles designated to receive Ticket escalation notifications.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
escalation_recipient_id	BIGINT	NO	AUTO_INCREMENT	Unique recipient identifier	PK	—
escalation_policy_id	BIGINT	NO	—	Related escalation policy	FK	escalation_policies.escalation_policy_id
user_id	BIGINT	YES	NULL	Specific user recipient	FK	users.user_id
role_id	BIGINT	YES	NULL	CRM role recipient	FK	roles.role_id

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
is_active	BOOLEAN	NO	TRUE	Whether recipient is active	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

The SRS explicitly requires escalation notification recipients to be configurable independently of RBAC roles.

17. ticket_escalation_history

Description: Maintains historical records of Ticket escalations, including escalations whose current flag has later cleared.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
escalation_history_id	BIGINT	NO	AUTO_INCREMENT	Unique escalation history identifier	PK	—
ticket_id	BIGINT	NO	—	Ticket that was escalated	FK	tickets.ticket_id
escalation_policy_id	BIGINT	YES	NULL	Policy that triggered escalation	FK	escalation_policies.escalation_policy_id
escalated_at	DATETIME	NO	CURRENT_TIMESTAMP	Escalation timestamp	—	—
cleared_at	DATETIME	YES	NULL	Timestamp escalation flag was cleared	—	—
inactivity_duration_minutes	INT	YES	NULL	Inactivity duration at escalation	—	—
notification_sent	BOOLEAN	NO	FALSE	Indicates whether notification was sent	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—

The SRS requires escalation history to be retained for reporting even after the live escalation flag clears.

18. ticket_comments

Description: Stores notes/comments and other recorded updates against Tickets.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
ticket_comment_id	BIGINT	NO	AUTO_INCREMENT	Unique comment identifier	PK	—
ticket_id	BIGINT	NO	—	Related Ticket	FK	tickets.ticket_id
user_id	BIGINT	NO	—	User who created comment	FK	users.user_id
comment_text	TEXT	NO	—	Comment or update content	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Comment timestamp	—	—

This supports the SRS rule that a Ticket's last-activity timestamp is updated by notes and other recorded activity.

19. programs

Description: Stores programs such as incubation, training, and workshops.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
program_id	BIGINT	NO	AUTO_INCREMENT	Unique Program identifier	PK	—
program_name	VARCHAR(255)	NO	—	Program name	—	—
program_type	VARCHAR(50)	NO	—	Incubation, training, workshop, etc.	—	—
description	TEXT	YES	NULL	Program description	—	—
start_date	DATE	YES	NULL	Program start date	—	—
end_date	DATE	YES	NULL	Program end date	—	—
is_active	BOOLEAN	NO	TRUE	Whether program is active	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

20. program_applications

Description: Represents a stakeholder's participation/application in a Program.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
program_application_id	BIGINT	NO	AUTO_INCREMENT	Unique Program Application identifier	PK	—
program_id	BIGINT	NO	—	Program being applied to	FK	programs.program_id
account_id	BIGINT	NO	—	Participating Account	FK	accounts.account_id
contact_id	BIGINT	YES	NULL	Participating Contact	FK	contacts.contact_id
opportunity_id	BIGINT	YES	NULL	Optional commercial Opportunity	FK	opportunities.opportunity_id
application_date	DATE	NO	CURRENT_DATE	Date application was submitted	—	—
status	VARCHAR(30)	NO	Applied	Application lifecycle status	Applied/Accepted/Enrolled/Completed/Withdrawn	—
progress_notes	TEXT	YES	NULL	Participant progress information	—	—
engagement_notes	TEXT	YES	NULL	Participant engagement information	—	—
completed_at	DATETIME	YES	NULL	Completion timestamp	—	—
withdrawn_at	DATETIME	YES	NULL	Withdrawal timestamp	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

The SRS explicitly introduces Program Application and defines its lifecycle as Applied → Accepted/Enrolled → Completed/Withdrawn.

21. events

Description: Stores events that can be scheduled and used for participant registration.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
event_id	BIGINT	NO	AUTO_INCREMENT	Unique Event identifier	PK	—
program_id	BIGINT	YES	NULL	Program associated with Event	FK	programs.program_id
event_name	VARCHAR(255)	NO	—	Event name	—	—
description	TEXT	YES	NULL	Event description	—	—
start_datetime	DATETIME	NO	—	Event start date/time	—	—
end_datetime	DATETIME	YES	NULL	Event end date/time	—	—
location	VARCHAR(255)	YES	NULL	Event location	—	—
capacity	INT	YES	NULL	Maximum participant capacity	—	—
status	VARCHAR(30)	NO	scheduled	Event status	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

22. event_registrations

Description: Stores participant registrations for Events.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
registration_id	BIGINT	NO	AUTO_INCREMENT	Unique registration identifier	PK	—
event_id	BIGINT	NO	—	Event being attended	FK	events.event_id
account_id	BIGINT	NO	—	Participating Account	FK	accounts.account_id
contact_id	BIGINT	YES	NULL	Registered Contact	FK	contacts.contact_id
registration_status	VARCHAR(30)	NO	registered	Registration status	—	—
registered_at	DATETIME	NO	CURRENT_TIMESTAMP	Registration timestamp	—	—
attended_at	DATETIME	YES	NULL	Attendance timestamp	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

23. notifications

Description: Stores system-generated notifications, including Ticket reminders and escalation notifications.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
notification_id	BIGINT	NO	AUTO_INCREMENT	Unique notification identifier	PK	—
recipient_user_id	BIGINT	YES	NULL	User receiving notification	FK	users.user_id
recipient_role_id	BIGINT	YES	NULL	Role receiving notification	FK	roles.role_id
ticket_id	BIGINT	YES	NULL	Related Ticket	FK	tickets.ticket_id
notification_type	VARCHAR(50)	NO	—	Reminder, escalation, system notification, etc.	—	—
title	VARCHAR(255)	NO	—	Notification title	—	—
message	TEXT	YES	NULL	Notification content	—	—
is_read	BOOLEAN	NO	FALSE	Whether notification was read	—	—
sent_at	DATETIME	YES	NULL	Time notification was sent	—	—
read_at	DATETIME	YES	NULL	Time notification was read	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—

For external escalation notifications, the SRS restricts content to Ticket ID, Priority, Category, inactivity duration, and a login-gated link.

24. audit_logs

Description: Stores audit history for security and field-level record changes.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
audit_log_id	BIGINT	NO	AUTO_INCREMENT	Unique audit record identifier	PK	—
user_id	BIGINT	YES	NULL	User who performed the action	FK	users.user_id
entity_type	VARCHAR(100)	NO	—	Entity/table affected	—	—
entity_id	BIGINT	NO	—	ID of affected record	—	—
action	VARCHAR(30)	NO	—	Create, update, delete, login, authorization, etc.	—	—
field_name	VARCHAR(100)	YES	NULL	Field that changed	—	—
old_value	TEXT	YES	NULL	Previous field value	—	—
new_value	TEXT	YES	NULL	New field value	—	—
ip_address	VARCHAR(45)	YES	NULL	Source IP address where applicable	—	—
user_agent	VARCHAR(500)	YES	NULL	Client information where applicable	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Audit timestamp	—	—

The SRS specifies field-level auditing for Lead, Opportunity, Account, Contact, and Ticket records.

25. export_records

Description: Tracks manual data exports performed for PMS handoff or reporting.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
export_record_id	BIGINT	NO	AUTO_INCREMENT	Unique export record identifier	PK	—
opportunity_id	BIGINT	YES	NULL	Closed Won Opportunity exported to PMS	FK	opportunities.opportunity_id
exported_by_user_id	BIGINT	NO	—	User who performed export	FK	users.user_id
export_type	VARCHAR(30)	NO	—	PMS handoff or report export	—	—
export_format	VARCHAR(20)	NO	—	CSV, Excel, or JSON	—	—
file_name	VARCHAR(255)	YES	NULL	Generated export file name	—	—
record_count	INT	YES	NULL	Number of records exported	—	—
status	VARCHAR(30)	NO	completed	Export status	—	—
exported_at	DATETIME	NO	CURRENT_TIMESTAMP	Export timestamp	—	—
error_message	TEXT	YES	NULL	Export error details if applicable	—	—

The PMS handoff is manual and supports CSV, Excel, and JSON.

26. reference_categories

Description: Stores configurable master/reference categories used by CRM modules.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
category_id	BIGINT	NO	AUTO_INCREMENT	Unique category identifier	PK	—
category_group	VARCHAR(50)	NO	—	Category group/type	—	—
category_name	VARCHAR(100)	NO	—	Category value	—	—
description	VARCHAR(255)	YES	NULL	Category description	—	—
is_active	BOOLEAN	NO	TRUE	Whether category is active	—	—
display_order	INT	YES	NULL	Display ordering	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

This table supports the SRS requirement that System Administrators maintain reference/master data.

27. business_hours_calendars

Description: Stores the configurable business-hours calendar required for Medium/Low Ticket escalation timing.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
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Column Name	Type (Size)	Null	Default	Description	Extra	Links To
calendar_id	BIGINT	NO	AUTO_INCREMENT	Unique calendar identifier	PK	—
calendar_name	VARCHAR(100)	NO	—	Calendar name	UNIQUE	—
timezone	VARCHAR(100)	YES	NULL	Calendar timezone	—	—
is_active	BOOLEAN	NO	TRUE	Whether calendar is active	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

28. business_hours

Description: Stores individual working-day hours for a business-hours calendar.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
business_hour_id	BIGINT	NO	AUTO_INCREMENT	Unique business-hours record	PK	—
calendar_id	BIGINT	NO	—	Business-hours calendar	FK	business_hours_calendars.calendar_id
day_of_week	TINYINT	NO	—	Day of week	1–7	—
start_time	TIME	YES	NULL	Start of business hours	—	—
end_time	TIME	YES	NULL	End of business hours	—	—
is_working_day	BOOLEAN	NO	TRUE	Whether day is a working day	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—
updated_at	DATETIME	NO	CURRENT_TIMESTAMP	Last update timestamp	—	—

The actual Ethiopian IT Park business-hours calendar remains an external dependency in SRS v1.0, so the database should keep these values configurable rather than hard-coding them.

29. business_holidays

Description: Stores holidays/exceptions for the configurable business-hours calendar.

Column Name	Type (Size)	Null	Default	Description	Extra	Links To
holiday_id	BIGINT	NO	AUTO_INCREMENT	Unique holiday identifier	PK	—
calendar_id	BIGINT	NO	—	Business-hours calendar	FK	business_hours_calendars.calendar_id
holiday_date	DATE	NO	—	Holiday date	—	—
holiday_name	VARCHAR(150)	NO	—	Holiday name	—	—
created_at	DATETIME	NO	CURRENT_TIMESTAMP	Creation timestamp	—	—